SOFTWARE REQUIREMENTS SPECIFICATION

One Step Coaching Platform

Version: 3.0 (Improved)

Date: April 11, 2026

Status: Final

Author: Software Requirements Analyst

Review: Based on Test Case Gap Assessment Analysis

TABLE OF CONTENTS

1. Introduction
2. Overall Description
3. System Features & Functional Requirements
4. External Interface Requirements
5. System Architecture
6. Database Design
7. Security Requirements
8. UI/UX Guidelines
9. Non-Functional Requirements
10. Testing Protocols
11. Use Cases
12. User Roles & Permissions
13. Appendices
14. Requirement-Test Traceability Matrix

---

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete requirements for the One Step Coaching Platform, a mobile-first learning management system (LMS) designed for exam preparation. This document serves as a comprehensive guide for developers, testers, project managers, and stakeholders involved in the system's development, deployment, and maintenance.

Improvement from v2.0: Added explicit reference to test case traceability and compliance requirements.

1.2 Scope

The One Step Coaching Platform enables:

Stakeholder	Capabilities
Students	Register, verify email, await admin approval, access learning content, take quizzes, view personal analytics
Teachers	Create/manage course content (subjects, chapters, topics), upload resources, create assessments, view class analytics
Administrators	Manage user approvals, assign roles, view system-wide analytics, manage user accounts

System provides: Secure authentication, role-based access control (RBAC), content management, quiz engine, analytics dashboard, profile management, file upload capabilities, and compliance with GDPR/CCPA data protection regulations.

Out of Scope: Mobile native applications, payment processing, third-party content licensing, live video conferencing.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
LMS	Learning Management System
RBAC	Role-Based Access Control
RLS	Row-Level Security (Supabase feature)
JWT	JSON Web Token
OAuth	Open Authorization protocol
API	Application Programming Interface
E2E	End-to-End
UI/UX	User Interface/User Experience
WCAG	Web Content Accessibility Guidelines
SSL/TLS	Secure Sockets Layer/Transport Layer Security
CRUD	Create, Read, Update, Delete
E2EE	End-to-End Encryption
CI/CD	Continuous Integration/Continuous Deployment
PWA	Progressive Web Application
OWASP	Open Web Application Security Project
GDPR	General Data Protection Regulation
CCPA	California Consumer Privacy Act
LCP	Largest Contentful Paint

Term	Definition
FID	First Input Delay
CLS	Cumulative Layout Shift
DR	Disaster Recovery

1.4 References

- IEEE 830 Standard for SRS Documents
- ISO/IEC/IEEE 29148 Systems and software engineering — Life cycle processes — Requirements engineering
- Supabase Documentation: https://supabase.com/docs
- Vue.js Documentation: https://vuejs.org/
- Chart.js Documentation: https://www.chartjs.org/
- OWASP Top 10 Security Vulnerabilities: https://owasp.org/www-project-top-ten/
- WCAG 2.1 Accessibility Guidelines: https://www.w3.org/WAI/WCAG21/quickref/
- GDPR Regulation: https://gdpr.eu/
- CCPA Regulation: https://oag.ca.gov/privacy/ccpa

1.5 Document Overview

This document is organized into 14 sections covering all aspects of the system from high-level overview to detailed technical specifications. Each section builds upon the previous to provide a complete understanding of system requirements.

Improvement from v2.0: Added Section 14 for Requirement-Test Traceability Matrix to ensure 100% requirement coverage.

---

2. OVERALL DESCRIPTION

2.1 Product Perspective

The One Step Coaching Platform is a standalone web application built using modern web technologies. It integrates with Supabase for backend services (authentication, database, storage) and is deployed as a Progressive Web Application (PWA) capable of being accessed from any device with a web browser.

System Context Diagram:

```
┌─────────────────────────────────────────────────────────────────────────┐
│                          One Step Coaching Platform                     │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────────┐          │
│  │   Vue.js     │  │ Vue Router   │  │   Chart.js           │
```

```
│   │   Frontend   □□□□  Navigation □  □   Analytics      □  □
│   └──────────────┘   └──────────────┘   └──────────────────────┘   │
│           │                                                        │
│           ▼                                                        │
│   ┌────────────────────────────────────────────────────────────┐   │
│   │                   Supabase Backend Services            □ □
│   │   ┌──────────┐   ┌──────────┐   ┌──────────────────┐       │   │
│   │   │    Auth   □  □ Database □  □    Storage      □    □ □
│   │   │ (GoTrue) □  □(PostgreSQL)□  □  (File Uploads) □    □ □
│   │   └──────────┘   └──────────┘   └──────────────────┘       │   │
│   └────────────────────────────────────────────────────────────┘   │
└────────────────────────────────────────────────────────────────────┘
          │                    │                    │
          ▼                    ▼                    ▼
┌──────────────┐   ┌──────────────────┐   ┌──────────────────┐
│   Email     □   □ Google OAuth  □   □  Web Browser  □
│   Service   □   □ (Optional)    □   □  (Any Device) □
└──────────────┘   └──────────────────┘   └──────────────────┘
```

2.2 Product Features

2.2.1 Core Features

Feature ID	Feature Name	Description	Priority	Test Coverage
FR-001	User Registration	Users can register with email/password or Google OAuth	High	AUTH-001 to AUTH-004
FR-002	Email Verification	Mandatory email verification before account activation	High	AUTH-008, UC-002
FR-003	Admin Approval Workflow	Admins must approve users before they can log in	High	AUTH-006, UC-003
FR-004	Role-Based Dashboards	Different UI/UX for Students, Teachers, and Admins	High	RBAC-001 to RBAC-009
FR-005	Content Management	Teachers can create subjects, chapters, topics, and content	High	CM-001 to CM-008
FR-006	File Upload	Support for PDF, images, and video links	Medium	CM-004 to CM-007
FR-007	Quiz Engine	Create, take, and score multiple-choice tests	High	QUIZ-001 to QUIZ-009
FR-008	Analytics Dashboard	Visual charts showing performance metrics	Medium	ANA-001 to ANA-005
FR-009	Profile Management	Users can update name, password, and profile picture	Medium	SEC-001 to SEC-006
FR-010	Google OAuth	Optional social login via Google	Medium	AUTH-005, OAuth-001

2.2.2 Secondary Features

Feature ID	Feature Name	Description	Priority	Test Coverage
FR-011	Password Recovery	Forgot password functionality via email	Medium	AUTH-013

Feature ID	Feature Name	Description	Priority	Test Coverage
FR-012	Session Management	Automatic token refresh and timeout handling	High	AUTH-009 to AUTH-010
FR-013	Responsive Design	Mobile-first design supporting all screen sizes	High	UI-001
FR-014	Accessibility	WCAG 2.1 AA compliance for screen readers	Medium	UI-005
FR-015	Multi-language Support	Support for 11+ languages (future enhancement)	Low	INT-006
FR-016	Data Export	Users can export their data (GDPR compliance)	High	GDPR-001
FR-017	Account Deletion	Users can request account deletion (Right to Erasure)	High	GDPR-002
FR-018	Real-time Notifications	WebSocket-based notifications for approvals/rejections	Medium	WS-001 to WS-003

Improvement from v2.0: Added FR-016, FR-017 for GDPR/CCPA compliance and FR-018 for real-time features with corresponding test coverage references.

2.3 User Classes and Characteristics

User Class	Description	Technical Skill Level	Frequency of Use
Student	Learners preparing for exams	Beginner to Intermediate	Daily
Teacher	Content creators and instructors	Intermediate	Multiple times daily
Administrator	System managers and approvers	Advanced	Multiple times daily
Guest	Unregistered visitors	Any	Occasional

2.4 Operating Environment

Component	Requirement
Browser	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
OS	Windows 10+, macOS 10.15+, Linux, iOS 14+, Android 10+
Network	HTTPS required, minimum 1 Mbps bandwidth
Screen Resolution	Minimum 320px width (mobile), 1024px (desktop)
JavaScript	ES6+ required, no legacy browser support

2.5 Design and Implementation Constraints

Constraint	Description	Verification Method
Technology Stack	Vue.js 3, Vite, Supabase, Chart.js	Code review
Budget	Free tier of Supabase (up to 500MB database, 1GB storage)	Resource monitoring
Timeline	MVP in 4 weeks, full features in 8 weeks	Project tracking
Compliance	GDPR, CCPA data protection regulations	Privacy audit
Security	All data encrypted in transit (TLS 1.3) and at rest	Security scan
Accessibility	WCAG 2.1 AA compliance	Automated + manual testing

2.6 Assumptions and Dependencies

Assumption	Impact if False	Mitigation
Users have valid email addresses	Registration will fail	Email format validation
Internet connectivity is available	App will show offline state	Offline mode with cached content
Supabase services remain operational	System will be unavailable	Health monitoring, alerting
Users accept cookies for session management	Sessions won't persist	Cookie consent banner
Third-party services (Google OAuth, Email) remain available	Social login/email delivery will fail	Fallback mechanisms, error handling

Dependency	Description	SLA
Supabase Auth	User authentication and session management	99.5% uptime
Supabase Database	PostgreSQL for all data storage	99.5% uptime
Supabase Storage	File uploads and media storage	99.5% uptime
Google OAuth API	Social login functionality	99.9% uptime
Netlify/Vercel	Hosting and deployment platform	99.9% uptime
Email Service	Verification and notification emails	99% delivery rate

---

3. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

3.1 User Registration (FR-001)

3.1.1 Description

Users can create new accounts by providing personal information and selecting their role.

3.1.2 Inputs

Field	Type	Required	Validation	Max Length
Full Name	Text	Yes	Non-empty, alphanumeric + spaces	100 characters
Mobile Number	Text	Yes	Format validation (international)	20 characters
Email Address	Email	Yes	RFC 5322 format, unique	254 characters
Password	Text	Yes	Min 8 chars, 1 letter, 1 number	128 characters
Role	Dropdown	Yes	Student or Teacher only	N/A

Boundary Value Testing:

- Full Name: 1 char, 50 chars, 99 chars, 100 chars, 101 chars, 500 chars
- Mobile: Valid US format, Valid international, No country code, Too many digits, Letters, Special characters
- Password: 7 chars (below min), 8 chars (min), 128 chars (max), 129 chars (above max), Only letters, Only numbers, Only special chars, Spaces at beginning/end

3.1.3 Processing Steps

Step	Action	System Response	Expected Result
1	Validate all input fields	Check format, length, uniqueness	All validations pass or specific error returned
2	Check email uniqueness in database	Query auth.users table	Email not found or error returned
3	Hash password using bcrypt	Supabase handles automatically	Password stored as hash
4	Create user in auth.users table	Supabase Auth creates record	User ID generated
5	Trigger handle_new_user()	Creates profile in public.users	Profile created with status="pending"
6	Send verification email via Supabase	Email queued for delivery	Email sent within 30 seconds
7	Set initial status to "pending"	Database update	Status visible in admin dashboard

3.1.4 Outputs

Output Type	Success	Failure
Message	"Registration successful. Please check your email for verification link."	Specific error message (duplicate email, invalid format, etc.)
Database	User record in auth.users, profile in public.users with status="pending"	No records created
Email	Verification email sent	No email sent

3.1.5 Business Rules

Rule ID	Rule Description	Enforcement Method
BR-001	Email must be unique across all users	Database unique constraint
BR-002	Password must be at least 8 characters	Client + server validation
BR-003	Password must contain at least 1 letter and 1 number	Regex validation
BR-004	Default role is "Student" if not specified	Default value in form
BR-005	Initial status is always "pending"	Trigger sets status
BR-006	Mobile number format must be valid	Regex pattern matching
BR-007	Full name cannot contain only whitespace	Trim + length validation

3.2 Email Verification (FR-002)

3.2.1 Description

Users must verify their email address before account activation.

3.2.2 Process Flow

User Registers Email Sent User Clicks Link Email Confirmed Status Remains Pending Admin Approval Required

3.2.3 Requirements

Requirement ID	Description	Verification Method
VER-001	Supabase GoTrue handles email verification automatically	System test
VER-002	Verification link expires after 24 hours	Time-based test
VER-003	Resend verification email option available	User interface test
VER-004	Unverified users cannot log in (enforced by Supabase Auth)	Authentication test
VER-005	Verification email delivered within 30 seconds	Performance test
VER-006	Verification link is single-use only	Security test

3.3 Admin Approval Workflow (FR-003)

3.3.1 Description

Administrators must approve user accounts before they can access the system.

3.3.2 Process Flow

User Verified Appears in Admin Dashboard Admin Reviews Approve/Reject Email Notification User Can Log In (if Approved)

3.3.3 Requirements

Requirement ID	Description	Expected Result
APR-001	Admin Dashboard displays all users filtered by status	Filtered list shows correctly
APR-002	Approve action updates status to "approved"	Status changes in database
APR-003	Reject action updates status to "rejected"	Status changes in database
APR-004	Email notification sent on approval/rejection	Email delivered within 30 seconds
APR-005	Manual database update possible as fallback	Admin can update via SQL
APR-006	Approval email includes welcome message and next steps	Email content verified
APR-007	Rejected users cannot re-register with same email without admin intervention	Database constraint enforced

3.4 User Login (FR-004)

3.4.1 Description

Authenticated users can log in using email/password or Google OAuth.

3.4.2 Preconditions

Condition	Verification
Email must be verified (email_confirmed_at != null)	Database check
Account status must be "approved"	Database check
Valid credentials provided	Authentication check

3.4.3 Processing Steps

Step	Action	System Response
1	Validate credentials with Supabase Auth	JWT token issued or error
2	Check email_confirmed_at is not null	Proceed or deny
3	Query public.users for status	Retrieve user record

Step	Action	System Response
4	If status != "approved", deny login with specific message	Error message displayed
5	If approved, issue JWT token and establish session	Session established

3.4.4 Error Messages

Condition	Error Message	HTTP Status
Invalid credentials	"Invalid email or password"	401
Email not verified	"Account not activated. Check your email for verification link."	403
Status pending	"Your registration is pending admin approval."	403
Status rejected	"Your registration was rejected. Contact support."	403

3.5 Role-Based Dashboards (FR-005)

3.5.1 Student Dashboard

Feature	Description	Restrictions
View available subjects and content	Browse learning materials	Cannot create or edit content
Navigate subject → chapter → topic hierarchy	Hierarchical navigation	Cannot access admin panel
View notes (PDF/text) and videos	Content consumption	Cannot view other students' results
Take quizzes and view results	Assessment participation	Cannot create tests
Access personal analytics	Performance tracking	Only own data visible

3.5.2 Teacher Dashboard

Feature	Description	Restrictions
Create and manage subjects	Content organization	Cannot approve/reject users
Create chapters and topics under subjects	Content structure	Cannot access admin panel
Upload content (PDF, video URLs, text notes)	Resource management	Cannot modify other teachers' content
Create tests with multiple-choice questions	Assessment creation	Cannot view system-wide analytics
View class analytics and student performance	Performance monitoring	Only own content's students visible
Manage own content only	CRUD operations	Limited to created content

3.5.3 Admin Dashboard

Feature	Description	Restrictions
View all users with filtering (pending/approved/rejected)	User management	None
Approve or reject user registrations	Account approval	None
Change user roles (Student ↔ Teacher ↔ Admin)	Role assignment	None
Delete users	Account removal	None
View system-wide analytics	Performance monitoring	None
Access all features of Teacher and Student roles	Full access	None

3.6 Content Management (FR-006)

3.6.1 Hierarchy Structure

Subject Chapter Topic Content Items

3.6.2 Content Types

Type	Description	Storage Method	Max Size
PDF Document	Files for download/viewing	Supabase Storage (coaching-assets bucket)	5MB
Video	Video links (YouTube, Vimeo)	URL stored in database	N/A
Text	Plain text notes	Stored in content_text column	10,000 characters

3.6.3 File Upload Requirements

Requirement	Specification
Maximum file size	2MB for profile pictures, 5MB for content
Allowed formats	PDF, JPG, PNG for content; JPG, PNG for profile
Storage location	Files stored in coaching-assets bucket
Access method	Signed URLs generated for private file access
Cleanup	Automatic cleanup on content deletion
Virus scanning	Files scanned before storage (security requirement)
File type validation	MIME type verification on upload

3.7 Quiz Engine (FR-007)

3.7.1 Test Creation

Input	Required	Validation	Default
Test title	Yes	Max 100 characters	N/A
Description	No	Max 500 characters	Empty
Duration in minutes	Yes	Min 1, Max 180	30
Passing score percentage	Yes	Min 0, Max 100	60%
Associated subject	Yes	Must exist	N/A
Active status	No	Boolean	true

Boundary Value Testing for Duration:

- 0 minutes (invalid)
- 1 minute (minimum)
- 179 minutes
- 180 minutes (maximum)
- 181 minutes (invalid)
- Decimal values (30.5 minutes - should reject)
- Negative values (should reject)

Boundary Value Testing for Passing Score:

- 0% (minimum)
- 59% (just below passing)
- 60% (passing threshold)
- 61% (just above passing)
- 100% (maximum)
- 101% (invalid)
- Negative values (should reject)

3.7.2 Question Creation

Input	Required	Validation	Default
Question text	Yes	Max 500 characters	N/A
Option A	Yes	Non-empty	N/A
Option B	Yes	Non-empty	N/A
Option C	Yes	Non-empty	N/A
Option D	Yes	Non-empty	N/A
Correct option	Yes	A, B, C, or D only	N/A
Points value	No	Positive integer	1

3.7.3 Test Taking Process

Step	Action	System Response
1	Student selects available test	Test questions displayed with timer
2	Student answers questions	Selections stored temporarily
3	Student clicks "Submit" or timer expires	Score calculated
4	System calculates score	Percentage computed
5	System saves attempt to database	Record created in test_attempts
6	System displays results	Score and pass/fail status shown

3.7.4 Scoring Logic

```
score = Σ(points for each correct answer)
total_points = Σ(all question points)
percentage = (score / total_points) × 100
passed = percentage >= passing_score
```

3.7.5 Timer Requirements

Requirement	Specification
Countdown timer displayed prominently	Visible in top-right corner
Auto-submit when timer reaches 0	Automatic submission triggered
Warning at 5 minutes remaining	Visual and auditory alert
Timer continues even if user navigates away	Session-based timer (server-side)
Timer synchronization	Server time as source of truth

3.8 Analytics Dashboard (FR-008)

3.8.1 Metrics Displayed

Metric	Calculation	Visualization	Access Control
Total Attempts	Count of test_attempts records	Number card	Student: Own, Teacher: Own tests, Admin: All
Average Score	Mean of all percentages	Number card	Student: Own, Teacher: Own tests, Admin: All
Pass Rate	(Passed / Total) × 100	Number card	Student: Own, Teacher: Own tests, Admin: All
Pass vs. Fail Distribution	Count by outcome	Doughnut Chart	Student: Own, Teacher: Own tests, Admin: All
Average Score per Test	Mean percentage grouped by test	Bar Chart	Student: Own, Teacher: Own tests, Admin: All
Performance Trend	Average score by date (last 7 days)	Line Chart	Student: Own, Teacher: Own tests, Admin: All

3.8.2 Access Control

Role	Data Visibility
Student	View only own results
Teacher	View results for tests they created
Admin	View all results system-wide

3.9 Profile Management (FR-009)

3.9.1 View Profile

Field	Editable	Description
Email	No	Read-only
Full Name	Yes	Editable, max 100 characters
Role	No	Read-only
Status	No	Read-only
Profile Picture	Yes	Upload/replace (max 2MB, JPG/PNG)
Account creation date	No	Read-only

3.9.2 Update Display Name

Step	Action	Validation
1	User enters new name	Non-empty, max 100 characters
2	System validates input	Trim whitespace, check length
3	Update public.users.full_name	Database update
4	Reflect immediately in UI	Frontend state update

3.9.3 Change Password

Step	Action	Validation
1	Enter current password	Verify against stored hash
2	Enter new password	Min 8 characters, 1 letter, 1 number
3	Confirm new password	Must match new password
4	Validate match	Comparison check

Step	Action	Validation
5	Update via Supabase Auth	Password hash updated
6	Force re-login on success	Session invalidated

3.9.4 Upload Profile Picture

Step	Action	Validation
1	Select image file	Max 2MB, JPG/PNG only
2	Upload to coaching-assets/profile/{user_id}/	Storage API call
3	Update auth.users metadata with avatar_url	Metadata update
4	Generate signed URL for display	Temporary access URL
5	Delete old image if replacing	Cleanup operation

3.10 GDPR/CCPA Compliance (FR-016, FR-017)

3.10.1 Data Export (Right to Data Portability)

Requirement	Specification
Export format	JSON or CSV
Exportable data	User profile, test attempts, content interactions
Export timeline	Within 30 days of request
Export method	Download link via email

3.10.2 Account Deletion (Right to Erasure)

Requirement	Specification
Deletion scope	All user data including profile, attempts, uploaded content
Deletion timeline	Within 30 days of request
Irreversible	Yes, with confirmation
Exception	Anonymized analytics data may be retained

---

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

4.1.1 Design Principles

Principle	Description
Mobile-First	Design for smallest screen first, scale up
Consistency	Uniform colors, fonts, and spacing throughout
Accessibility	WCAG 2.1 AA compliance
Feedback	Immediate visual feedback for all actions
Error Prevention	Validation before submission
Forgiveness	Easy undo/rollback for destructive actions

4.1.2 Color Palette

Color	Hex Code	Usage
Primary Green	#2E7D32	Primary buttons, headers, success states
Primary Dark	#1B5E20	Hover states, gradients
Secondary Blue	#1976D2	Secondary buttons, links
Accent Amber	#FFC107	Highlights, warnings, CTAs

Color	Hex Code	Usage
Danger Red	#D32F2F	Error states, delete actions
Background	#F5F7FA	Page backgrounds
Card Background	#FFFFFF	Cards, modals
Text Dark	#1A202C	Primary text
Text Muted	#718096	Secondary text
Border	#E2E8F0	Input borders, dividers

4.1.3 Typography

Element	Font Family	Size	Weight
Headings (H1-H6)	Segoe UI, sans-serif	32px-14px	700
Body Text	Segoe UI, sans-serif	16px	400
Buttons	Segoe UI, sans-serif	16px	600
Small Text	Segoe UI, sans-serif	14px	400
Code	Monospace	14px	400

4.2 Hardware Interfaces

Device Type	Requirements
Desktop	Keyboard, mouse, 1024px+ width
Tablet	Touch input, 768px+ width
Mobile	Touch input, 320px+ width, camera (for profile pic)

4.3 Software Interfaces

Interface	Purpose	Protocol	SLA
Supabase Auth	User authentication	REST API	99.5%
Supabase Database	Data persistence	PostgreSQL	99.5%
Supabase Storage	File uploads	REST API	99.5%
Google OAuth	Social login	OAuth 2.0	99.9%
Email Service	Verification emails	SMTP (Supabase)	99%
Hosting (Netlify/Vercel)	Deployment	HTTPS	99.9%

4.4 Communication Interfaces

Protocol	Port	Encryption	Purpose
HTTPS	443	TLS 1.3	All web traffic
WebSocket	443	WSS	Real-time notifications

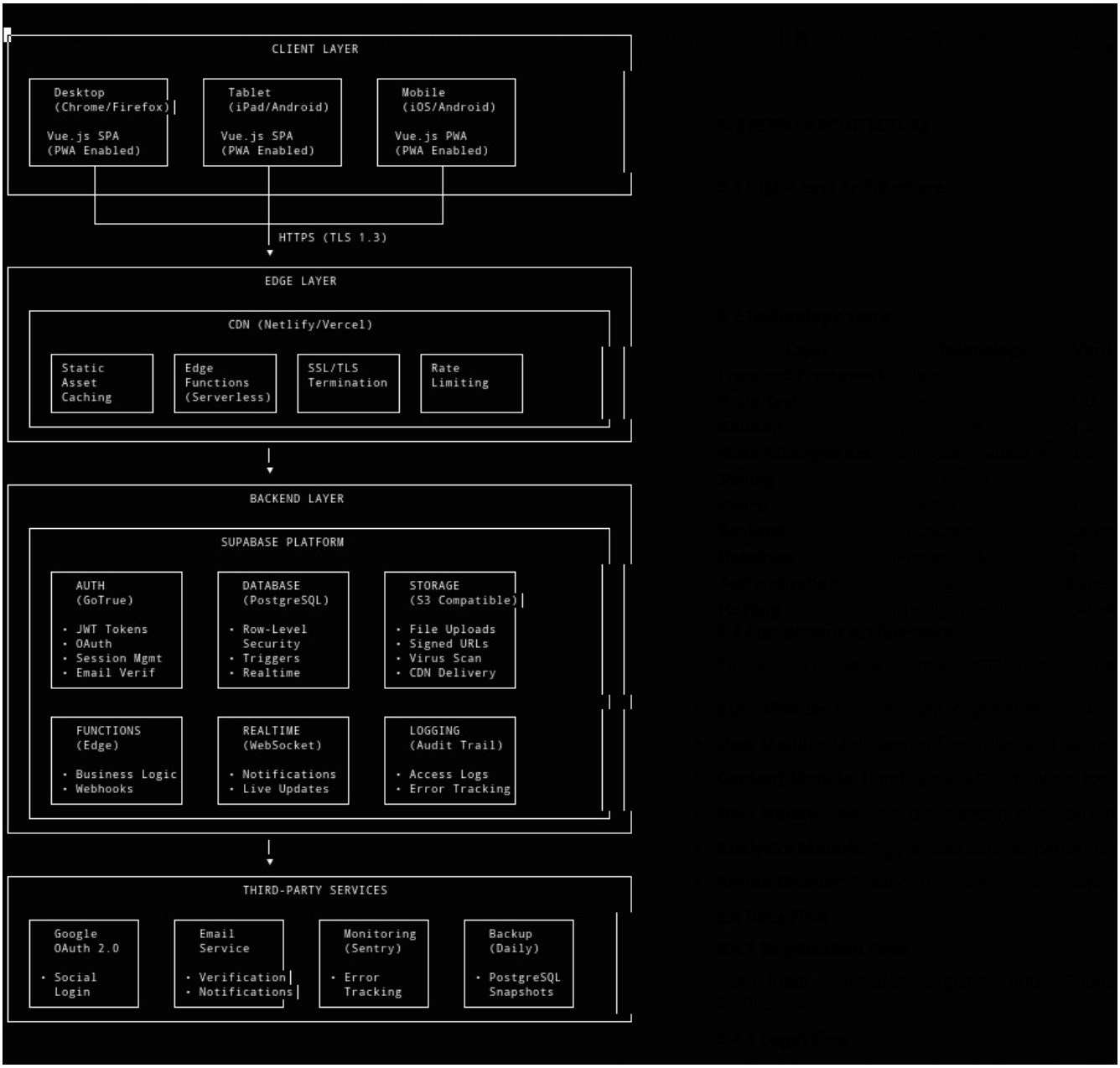
Security Requirements:

- TLS 1.3 certificate validation on all connections
- Cipher suite verification (AES-256-GCM minimum)
- Certificate pinning for mobile clients (future enhancement)

---

5. SYSTEM ARCHITECTURE

5.1 High-Level Architecture



5.2 Technology Stack

Layer	Technology	Version	Purpose
Frontend Framework	Vue.js	3.4.x	Reactive UI components
Build Tool	Vite	5.0.x	Fast bundling and dev server
Routing	Vue Router	4.2.x	Client-side navigation

Layer	Technology	Version	Purpose
State Management	Vue Composition API	3.x	Component state logic
Styling	CSS3 + SCSS	-	Responsive design
Charts	Chart.js	4.4.x	Data visualization
Backend	Supabase	Latest	Auth, DB, Storage
Database	PostgreSQL	14+	Relational data storage
Authentication	GoTrue	Latest	JWT-based auth
Hosting	Netlify/Vercel	Latest	Static site hosting & Edge functions

5.3 Component Architecture

The system follows a modular component structure:

- **Auth Module**: Handles login, registration, session management, and OAuth flows.
- **User Module**: Manages profiles, roles, and approval workflows.
- **Content Module**: Handles subjects, chapters, topics, and file uploads.
- **Quiz Module**: Manages test creation, question banks, timer logic, and scoring.
- **Analytics Module**: Aggregates data for performance visualization.
- **Admin Module**: Provides user management and system oversight.

5.4 Data Flow

5.4.1 Registration Flow

Client Input Validation Supabase Auth Trigger (handle_new_user) Public Users Table Email Verification

5.4.2 Login Flow

Credentials Supabase Auth Check Email Verified Check Status Approved Issue JWT Set Session

5.4.3 Content Upload Flow

File Select Client Validation (Size/Type) Signed URL Generation Direct Upload to Storage DB Record Update

---

6. DATABASE DESIGN

6.1 Entity Relationship Diagram (Conceptual)

```
[auth.users] 1:1 [public.users]
[public.users] 1:N [public.subjects] (created_by)
[public.subjects] 1:N [public.chapters]
[public.chapters] 1:N [public.topics]
[public.topics] 1:N [public.content_items]
[public.subjects] 1:N [public.tests]
[public.tests] 1:N [public.questions]
[public.users] 1:N [public.test_attempts]
```

[public.tests] 1:N [public.test_attempts]

6.2 Table Specifications

6.2.1 auth.users (Supabase Managed)

Managed by Supabase Auth; direct writes restricted.

Column	Type	Constraints	Description
id	UUID	PK	Unique user identifier
email	CITEXT	UNIQUE, NOT NULL	User email address
encrypted_password	TEXT	NOT NULL	Bcrypt hashed password
email_confirmed_at	TIMESTAMPTZ	NULLABLE	Verification timestamp
created_at	TIMESTAMPTZ	DEFAULT NOW()	Account creation time
raw_user_meta_data	JSONB	NULLABLE	Custom metadata (name, role, etc.)

6.2.2 public.users

Column	Type	Constraints	Description
id	UUID	PK, FK → auth.users	Links to auth user
full_name	TEXT	NOT NULL	Display name
mobile_number	TEXT	NOT NULL	Contact number
role	TEXT	NOT NULL, CHECK IN ('Student','Teacher','Admin')	User role
status	TEXT	NOT NULL, CHECK IN ('pending','approved','rejected')	Approval status
avatar_url	TEXT	NULLABLE	Profile picture path
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last update time

6.2.3 public.subjects

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique subject ID
name	TEXT	NOT NULL	Subject name
description	TEXT	NULLABLE	Subject description
created_by	UUID	FK → public.users	Teacher who created
created_at	TIMESTAMPTZ	DEFAULT NOW()	Creation timestamp

6.2.4 public.chapters

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique chapter ID
subject_id	UUID	FK → public.subjects, NOT NULL	Parent subject
title	TEXT	NOT NULL	Chapter title
description	TEXT	NULLABLE	Chapter description
order_index	INT	DEFAULT 0	Display order
created_at	TIMESTAMPTZ	DEFAULT NOW()	Creation timestamp

6.2.5 public.topics

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique topic ID
chapter_id	UUID	FK → public.chapters, NOT NULL	Parent chapter
title	TEXT	NOT NULL	Topic title
description	TEXT	NULLABLE	Topic description
order_index	INT	DEFAULT 0	Display order
created_at	TIMESTAMPTZ	DEFAULT NOW()	Creation timestamp

6.2.6 public.content_items

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique content ID
topic_id	UUID	FK → public.topics, NOT NULL	Parent topic
type	TEXT	NOT NULL, CHECK IN ('pdf','video','text')	Content type
title	TEXT	NOT NULL	Content title
file_url	TEXT	NULLABLE	Storage path for PDFs/images
video_url	TEXT	NULLABLE	External video link
content_text	TEXT	NULLABLE	Plain text content
created_at	TIMESTAMPTZ	DEFAULT NOW()	Creation timestamp

6.2.7 public.tests

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique test ID
subject_id	UUID	FK → public.subjects, NULLABLE	Associated subject
title	TEXT	NOT NULL	Test title
description	TEXT	NULLABLE	Test description
duration_minutes	INT	DEFAULT 30, CHECK > 0	Time limit
passing_score	INT	DEFAULT 60, CHECK 0-100	Passing percentage
created_by	UUID	FK → public.users, NOT NULL	Teacher creator
is_active	BOOLEAN	DEFAULT TRUE	Published status
created_at	TIMESTAMPTZ	DEFAULT NOW()	Creation timestamp

6.2.8 public.questions

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique question ID
test_id	UUID	FK → public.tests, NOT NULL	Parent test
question_text	TEXT	NOT NULL	Question content
option_a	TEXT	NOT NULL	First option
option_b	TEXT	NOT NULL	Second option
option_c	TEXT	NOT NULL	Third option
option_d	TEXT	NOT NULL	Fourth option
correct_option	CHAR(1)	NOT NULL, CHECK IN ('A','B','C','D')	Correct answer
points	INT	DEFAULT 1, CHECK > 0	Question weight
created_at	TIMESTAMPTZ	DEFAULT NOW()	Creation timestamp

6.2.9 public.test_attempts

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique attempt ID
test_id	UUID	FK → public.tests, NOT NULL	Test taken
user_id	UUID	FK → public.users, NOT NULL	Student who took it
score	INT	NOT NULL	Points earned
total_points	INT	NOT NULL	Maximum possible
percentage	NUMERIC(5,2)	NOT NULL	Calculated percentage
answers	JSONB	NOT NULL	{question_id: selected_option}
started_at	TIMESTAMPTZ	DEFAULT NOW()	Test start time
completed_at	TIMESTAMPTZ	NULLABLE	Test end time

6.3 Indexes

Index Name	Table	Columns	Purpose
idx_users_status	public.users	status	Fast filtering by approval status
idx_users_role	public.users	role	Fast role-based queries
idx_chapters_subject	public.chapters	subject_id	Quick chapter retrieval
idx_topics_chapter	public.topics	chapter_id	Quick topic retrieval
idx_content_topic	public.content_items	topic_id	Content lookup
idx_questions_test	public.questions	test_id	Question retrieval
idx_tests_subject	public.tests	subject_id	Subject-based tests
idx_attempts_user	public.test_attempts	user_id	Student's attempts
idx_attempts_test	public.test_attempts	test_id	Test results
idx_attempts_completed	public.test_attempts	completed_at	Recent attempts sorting

6.4 Triggers

6.4.1 handle_new_user()

Creates a profile row in public.users whenever a new user signs up in auth.users.

```
CREATE OR REPLACE FUNCTION public.handle_new_user()
RETURNS TRIGGER AS $$
BEGIN
  INSERT INTO public.users (id, full_name, mobile_number, role, status)
  VALUES (
    NEW.id,
    COALESCE(NEW.raw_user_meta_data->>'full_name', ''),
    COALESCE(NEW.raw_user_meta_data->>'mobile_number', ''),
    COALESCE(NEW.raw_user_meta_data->>'role', 'Student'),
    'pending'
  );
```

```
    RETURN NEW;
END;
$$ LANGUAGE plpgsql SECURITY DEFINER;

CREATE TRIGGER on_auth_user_created
AFTER INSERT ON auth.users
FOR EACH ROW EXECUTE FUNCTION public.handle_new_user();
```

6.4.2 update_updated_at()

Automatically updates the updated_at timestamp on record modification.

```
CREATE OR REPLACE FUNCTION update_updated_at_column()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = NOW();
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER update_users_updated_at
BEFORE UPDATE ON public.users
FOR EACH ROW EXECUTE FUNCTION update_updated_at_column();
```

---

7. SECURITY REQUIREMENTS

7.1 Authentication Security

7.1.1 Password Requirements

Requirement	Specification
Minimum Length	8 characters
Character Types	At least 1 letter and 1 number
Hashing Algorithm	bcrypt (handled by Supabase GoTrue)
Storage	Never stored in plaintext
Transmission	TLS 1.3 encrypted

7.1.2 Session Management

Aspect	Specification
Token Type	JWT (JSON Web Token)

Aspect	Specification
Expiration	1 hour access token, 7 day refresh token
Auto-refresh	Silent refresh before expiration
Storage	HttpOnly cookies (preferred) or secure localStorage
Logout	Server-side token revocation

7.1.3 OAuth Security

Aspect	Specification
Provider	Google OAuth 2.0
Scopes	email, profile
Redirect URI	Supabase callback endpoint only
State Parameter	CSRF protection enabled
PKCE	Enabled for public clients

7.2 Row-Level Security (RLS) Policies

Enforcement: All tables in public schema must have RLS enabled.

7.2.1 public.users Table

Policy Name	Operation	Condition
Users can view own profile	SELECT	auth.uid() = id
Admins can view all users	SELECT	EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role = 'Admin')
Users can update own profile	UPDATE	auth.uid() = id
Admins can update all users	UPDATE	EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role = 'Admin')
System can insert profiles	INSERT	Trigger-based (handle_new_user)

7.2.2 public.subjects Table

Policy Name	Operation	Condition
Public view subjects	SELECT	TRUE (all authenticated users)
Teachers create subjects	INSERT	EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role IN ('Teacher', 'Admin'))
Teachers manage own subjects	UPDATE/ DELETE	created_by = auth.uid()

7.2.3 public.tests Table

Policy Name	Operation	Condition
Students view active tests	SELECT	is_active = TRUE AND EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role IN ('Student', 'Teacher', 'Admin'))
Teachers create tests	INSERT	EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role IN ('Teacher', 'Admin'))

Policy Name	Operation	Condition
Teachers manage own tests	UPDATE/ DELETE	created_by = auth.uid()

7.2.4 public.test_attempts Table

Policy Name	Operation	Condition
Students view own attempts	SELECT	user_id = auth.uid()
Teachers view test attempts	SELECT	EXISTS (SELECT 1 FROM public.tests t WHERE t.id = test_attempts.test_id AND t.created_by = auth.uid()) OR EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role = 'Admin')
Insert attempts	INSERT	user_id = auth.uid() OR EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role = 'Admin')

7.3 Storage Security

7.3.1 Bucket Configuration

Setting	Value
Bucket Name	coaching-assets
Public Access	FALSE (Private)
File Size Limit	5MB per file
Allowed MIME Types	image/jpeg, image/png, application/pdf

7.3.2 Storage Policies

Policy Name	Operation	Condition
Users can upload profile pics	INSERT	bucket_id = 'coaching-assets' AND (storage.foldername(name))[1] = auth.uid()::text
Users can view own profile pics	SELECT	bucket_id = 'coaching-assets' AND (storage.foldername(name))[1] = auth.uid()::text
Teachers can upload content	INSERT	bucket_id = 'coaching-assets' AND (storage.foldername(name))[1] = 'content' AND EXISTS (SELECT 1 FROM public.users WHERE id = auth.uid() AND role IN ('Teacher', 'Admin'))
Everyone can view content	SELECT	bucket_id = 'coaching-assets' AND (storage.foldername(name))[1] = 'content'

7.4 Network Security

Requirement	Specification
HTTPS	Mandatory (TLS 1.3)
CORS	Restricted to approved domains only
Rate Limiting	100 requests/minute per IP (Supabase default)
Input Validation	All inputs sanitized on client and server
SQL Injection	Prevented via parameterized queries (Supabase ORM)
XSS Prevention	Vue.js auto-escapes by default

7.5 Data Protection & Compliance

Aspect	Specification
Data at Rest	Encrypted (Supabase PostgreSQL encryption)
Data in Transit	Encrypted (TLS 1.3)
Backup	Daily automated backups (Supabase managed)
Retention	User data retained until account deletion
GDPR Compliance	Right to erasure supported via account deletion
CCPA Compliance	Data export and deletion rights supported

---

8. UI/UX GUIDELINES

8.1 Design Principles

1. **Clarity**: Interface elements should be self-explanatory.
2. **Consistency**: Uniform patterns across all pages.
3. **Efficiency**: Minimize clicks to complete common tasks.
4. **Feedback**: Immediate response to all user actions.
5. **Forgiveness**: Easy undo/rollback for destructive actions.

8.2 Accessibility (WCAG 2.1 AA)

Guideline	Implementation
Color Contrast	Minimum 4.5:1 for normal text
Keyboard Navigation	All interactive elements focusable
Screen Readers	ARIA labels on all form elements
Text Scaling	Support up to 200% zoom
Error Identification	Clear error messages with suggestions

8.3 Layout Specifications

8.3.1 Grid System

- **Desktop (≥1024px)**: 12-column grid, 24px gutters
- **Tablet (768px-1023px)**: 8-column grid, 16px gutters
- **Mobile (≤767px)**: 4-column grid, 12px gutters

8.3.2 Spacing Scale

Unit	Pixel Value	Usage
xs	4px	Tight spacing
sm	8px	Compact spacing
md	16px	Standard spacing
lg	24px	Section spacing
xl	32px	Major spacing
xxl	48px	Page margins

8.4 Error Handling

Type	Background	Border	Icon
Error	#FFF5F5	#D32F2F	⚠️
Warning	#FFFBEA	#FFA000	⚡
Success	#F0FFF4	#2E7D32	✅
Info	#EBF8FF	#1976D2	ℹ️

8.5 Responsive Behavior

- **Mobile**: Hamburger menu, full-width buttons, stacked forms, touch targets □44px.
- **Desktop**: Hover states, multi-column layouts, keyboard shortcuts, drag-and-drop.

---

9. NON-FUNCTIONAL REQUIREMENTS

9.1 Performance Requirements

Metric	Target	Measurement Method
Page Load Time	< 3 seconds	Lighthouse audit
First Contentful Paint	< 1.5 seconds	Chrome DevTools
Time to Interactive	< 3.5 seconds	Lighthouse audit
API Response Time	< 500ms (p95)	Network tab monitoring
Database Query Time	< 100ms (p95)	PostgreSQL EXPLAIN ANALYZE
File Upload Speed	< 5 seconds (5MB)	Manual testing
Concurrent Users	100+ users	Load testing

Core Web Vitals Targets:

- **LCP (Largest Contentful Paint)**: < 2.5 seconds
- **FID (First Input Delay)**: < 100 milliseconds
- **CLS (Cumulative Layout Shift)**: < 0.1

9.2 Scalability Requirements

Aspect	Current Capacity	Future Target
Database Size	500MB (Free Tier)	10GB (Pro Plan)
Storage	1GB (Free Tier)	100GB (Pro Plan)
Monthly Active Users	10,000	100,000
API Requests/Day	100,000	1,000,000
Concurrent Connections	500	5,000

9.3 Reliability Requirements

Metric	Target	Recovery Time
Uptime	99.5%	< 4 hours/month downtime
Data Loss	0%	Daily automated backups
Disaster Recovery	< 24 hours	Restore from backup
Error Rate	< 1%	Monitoring alerts

9.4 Maintainability Requirements

Aspect	Standard
Code Coverage	> 80% unit tests
Documentation	Inline comments + README
Version Control	Git with semantic versioning
CI/CD	Automated testing on push
Logging	Structured logging (JSON format)
Monitoring	Error tracking (Sentry) + Analytics

9.5 Compatibility Requirements

Platform	Minimum Version	Support Level
Chrome	90+	Full
Firefox	88+	Full
Safari	14+	Full
Edge	90+	Full
iOS Safari	14+	Full
Chrome Android	90+	Full

---

10. TESTING PROTOCOLS

10.1 Testing Strategy

Adopt a **Testing Pyramid** approach:

1. **Unit Tests (70%)**: Vitest + Vue Test Utils (Run on every commit)
2. **Integration Tests (20%)**: Supabase Test DB + API testing (Run on PR)
3. **E2E Tests (10%)**: Cypress (Run on staging deployment)
4. **Performance/Security**: Scheduled weekly and on every build respectively.

10.2 Test Cases Summary

10.2.1 Authentication Testing

Test ID	Description	Expected Result	Priority
AUTH-001	Register with valid email/password	Success, verification email sent	High
AUTH-002	Register with invalid email format	Error: "Invalid email format"	High
AUTH-003	Register with password < 8 chars	Error: "Password too short"	High
AUTH-004	Register with duplicate email	Error: "Email already exists"	High
AUTH-005	Login with correct credentials	Success, redirect to dashboard	High
AUTH-006	Login with pending user	Error: "Pending approval"	High
AUTH-007	Login with rejected user	Error: "Registration rejected"	High
AUTH-008	Login with unverified email	Error: "Verify email first"	High
AUTH-009	Session expires after timeout	Redirect to login	Medium
AUTH-010	Refresh token works	Session extended silently	Medium
AUTH-011	Logout functionality	Clear session, redirect home	High

Test ID	Description	Expected Result	Priority
AUTH-012	Access protected route without login	Redirect to login	High
AUTH-013	Password recovery flow	Reset email sent, link valid	Medium

10.2.2 Role-Based Access Control (RBAC)

Test ID	Description	Expected Result	Priority
RBAC-001	Student accesses admin panel	Redirect to student dashboard	High
RBAC-002	Student creates content	Error: "Access denied"	High
RBAC-003	Student views content	Success	High
RBAC-004	Teacher approves users	Error: "Access denied"	High
RBAC-005	Teacher creates subject	Success	High
RBAC-006	Teacher edits other's content	Error: "Access denied"	High
RBAC-007	Admin accesses all features	Success	High
RBAC-008	Admin approves user	User status changes to approved	High
RBAC-009	Admin changes user role	Role updated successfully	Medium

10.2.3 Content Management

Test ID	Description	Expected Result	Priority
CM-001	Create subject	Success, appears in list	High
CM-002	Create chapter under subject	Success, linked correctly	High
CM-003	Create topic under chapter	Success, linked correctly	High
CM-004	Upload PDF content	Success, file stored	High
CM-005	Add video URL	Success, displays correctly	High
CM-006	Upload file > 5MB	Error: "File too large"	Medium
CM-007	Unauthorized content upload	Error: "Access denied"	High
CM-008	Delete content	Success, removed from DB	Medium

10.2.4 Quiz System

Test ID	Description	Expected Result	Priority
QUIZ-001	Create test with valid data	Success	High
QUIZ-002	Create test with missing questions	Error: "Add at least 1 question"	Medium
QUIZ-003	Student takes test	Success, timer works	High
QUIZ-004	Timer expires during test	Auto-submit, score calculated	High
QUIZ-005	Submit test with all correct	Score = 100%	High
QUIZ-006	Submit test with all wrong	Score = 0%	High
QUIZ-007	Score stored in database	Record created in test_attempts	High
QUIZ-008	Edit test after creation	Success, changes saved	Medium
QUIZ-009	Delete test	Success, questions cascade delete	Medium

10.2.5 Security Testing (OWASP Top 10 Focus)

Test ID	Description	Expected Result	Priority
SEC-001	Unauthorized API access	401/403 error	High
SEC-002	JWT tampering	Request rejected	High
SEC-003	SQL injection attempt	Query fails safely	High
SEC-004	XSS attempt in form fields	Input escaped	High

Test ID	Description	Expected Result	Priority
SEC-005	Rate limiting exceeded	429 Too Many Requests	Medium
SEC-006	File upload security	Malicious files blocked	High
SEC-007	Brute force attack simulation	Account lockout after threshold	Critical
SEC-008	Session fixation attack	New session token issued	Critical
SEC-009	CSRF on state-changing ops	Request rejected	Critical
SEC-010	IDOR (Insecure Direct Object Ref)	RLS policy blocks access	Critical

10.2.6 Performance Testing

Test ID	Description	Expected Result	Priority
PERF-001	Page load < 3 seconds	Passes Lighthouse	High
PERF-002	API response < 500ms	Meets target	High
PERF-003	DB query < 100ms	With indexes	High
PERF-004	Load test 100 users	No errors	Medium
PERF-005	Image loading speed	< 2 seconds	Medium
PERF-006	Stress test (500+ users)	System degrades gracefully	Medium

10.3 Testing Environment

Environment	Purpose	URL
Development	Local testing	localhost:5173
Staging	Pre-production testing	staging.one-step-coaching.app
Production	Live application	one-step-coaching.app

10.4 Defect Management

Severity	Definition	Response Time
Critical	System down, data loss	24 hours
High	Major feature broken	48 hours
Medium	Minor feature issue	1 week
Low	Cosmetic, enhancement	Next sprint

---

11. USE CASES

11.1 UC-001: User Registration

- **Actor**: Guest
- **Precondition**: User has valid email address
- **Postcondition**: User account created with "pending" status
- **Main Flow**:

1. User navigates to registration page.
2. User fills in: Full Name, Mobile, Email, Password, Role.
3. User clicks "Register".
4. System validates all inputs.
5. System creates auth user in Supabase.

6. System triggers profile creation in public.users.
7. System sends verification email.
8. System displays success message.

- **Alternative Flows**:

1. A1: Invalid email format Show error, prevent submission.
2. A2: Duplicate email Show error, prevent submission.
3. A3: Weak password Show error, prevent submission.

11.2 UC-002: Email Verification

- **Actor**: Registered User
- **Precondition**: User has registered, verification email sent
- **Postcondition**: Email confirmed, account ready for approval
- **Main Flow**:

1. User receives verification email.
2. User clicks verification link.
3. System marks email as confirmed (email_confirmed_at set).
4. System displays confirmation page.
5. User status remains "pending" (awaiting admin approval).

11.3 UC-003: Admin Approval

- **Actor**: Administrator
- **Precondition**: User has verified email, status = "pending"
- **Postcondition**: User status = "approved", can log in
- **Main Flow**:

1. Admin logs in.
2. Admin navigates to Admin Dashboard.
3. Admin views pending users list.
4. Admin clicks "Approve" on user.
5. System updates status to "approved".
6. System sends approval email to user.
7. User can now log in.

11.4 UC-004: Student Takes Test

- **Actor**: Student
- **Precondition**: Student is approved, test is active

- **Postcondition**: Test attempt recorded, score calculated
- **Main Flow**:

1. Student navigates to "Take Tests".
2. Student selects available test.
3. System displays test questions with timer.
4. Student answers questions.
5. Student clicks "Submit".
6. System calculates score.
7. System saves attempt to database.
8. System displays results.

- **Alternative Flows**:

1. A1: Timer expires Auto-submit, score calculated.
2. A2: Network disconnect Show warning, retry on reconnect.

11.5 UC-005: Teacher Creates Content

- **Actor**: Teacher
- **Precondition**: Teacher is approved
- **Postcondition**: Content available to students
- **Main Flow**:

1. Teacher logs in.
2. Teacher navigates to Dashboard Manage Content.
3. Teacher selects subject.
4. Teacher creates chapter.
5. Teacher creates topic.
6. Teacher uploads content (PDF/video/text).
7. System saves content to database.
8. Content appears in student view.

---

12. USER ROLES & PERMISSIONS

12.1 Role Matrix

Permission	Student	Teacher	Admin
Register Account	✓	✓	✓
Verify Email	✓	✓	✓

Permission	Student	Teacher	Admin
Login (if approved)	✓	✓	✓
View Profile	✓	✓	✓
Update Profile Name	✓	✓	✓
Change Password	✓	✓	✓
Upload Profile Picture	✓	✓	✓
View Subjects	✓	✓	✓
View Content	✓	✓	✓
Take Tests	✓	✓	✓
View Own Results	✓	✓	✓
Create Subjects	✗	✓	✓
Create Chapters	✗	✓	✓
Create Topics	✗	✓	✓
Upload Content	✗	✓	✓
Create Tests	✗	✓	✓
Create Questions	✗	✓	✓
View Class Analytics	✗	✓	✓
View System Analytics	✗	✗	✓
View All Users	✗	✗	✓
Approve Users	✗	✗	✓
Reject Users	✗	✗	✓
Change User Roles	✗	✗	✓
Delete Users	✗	✗	✓
Delete Own Content	✗	✓	✓
Delete Other's Content	✗	✗	✓
Access Admin Panel	✗	✗	✓

12.2 Role Descriptions

- **Student**: Primary goal is to learn and pass exams. Read-only for content, write access only for test attempts and profile.
- **Teacher**: Primary goal is to create and manage content. Full CRUD on own content. Read access to class analytics.
- **Administrator**: Primary goal is system management. Full system access. Expected to act responsibly.
- **Guest**: Unauthenticated visitor. Read-only access to Home page.

---

13. APPENDICES

13.1 Glossary of Terms

- **Supabase**: Open-source Firebase alternative providing Auth, DB, Storage.
- **GoTrue**: The authentication engine used by Supabase.

- **RLS**: Row-Level Security: Database policies restricting row access based on user identity.
- **JWT**: JSON Web Token: Compact means to represent claims between parties.
- **OAuth 2.0**: Open standard for access delegation (social login).
- **PWA**: Progressive Web App: App-like experience via web technologies.
- **CI/CD**: Continuous Integration/Deployment: Automated testing and deployment.
- **ESLint**: Tool for identifying and fixing JavaScript problems.
- **Vite**: Next-generation frontend build tooling.

13.2 Deployment Checklist

13.2.1 Pre-Deployment

- All unit tests passing (>80% coverage)
- All E2E tests passing
- Security audit completed (no critical vulnerabilities)
- Environment variables configured in hosting platform